

newtxsf—A Sans Serif Math Package

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1 Introduction

I was spurred to make this package by one of the excellent features of the STIX fonts—mathematical Roman and Greek sans serif alphabets with all glyphs unmistakable, unlike most sans serif fonts. These are sized to match Times, with xheights near 440em, rather smaller than most sans serif fonts. So, this package was constructed by replacing the existing math italic in newtxmath with the sans serif glyphs from STIX and simplifying the options. The result is a math package that blends well with sans serif text fonts with an xheight similar to that of Times. Surprisingly, it does seem to work well with some other sans serif fonts with larger xheights, such as *FiraSans* at about 530em. With modern-day projectors having considerably higher resolutions than a few years ago, large xheight may be less of a concern than it once was. (Note: *FiraSans* also works well with the *Arev Sans* package, where the xheight is larger.)

2 Package Options

The options are much simpler than in newtxmath, as I have ruled out a number which, in my opinion, were of interest only for backward compatibility. These are the options you may specify.

- `scaled` allows you to scale all fonts in this math package to match a chosen text font family.
- `nosymbolsc` saves you a math group of mostly rather obscure symbols.
- `cmintegrals` makes use of integrals drawn from Computer Modern rather than the more upright, but unattractive, `txfonts` integrals. (This option no longer has any effect since newtx now uses a similar collection of integral signs.)
- `upint` produces upright integral signs. (The default shape is slanted.)
- `noamssymbols` saves you one or two math groups (the AMS symbols) if you have no need of them. If you use an AMS document class, like `amsart`, be sure to specify `noamsfonts` among its options, if using a recent version of the AMS class.
- `amssymbols` does the opposite—this is the default.
- `uprightGreek` specifies the use of upright rather than slanted Greek symbols for uppercase only.
- `slantedGreek` does the opposite.
- `frenchmath` forces uppercase and lowercase Greek to upright shape and makes uppercase math Roman letters render in upright rather than slanted shape.
- `noSTIXops` (added in version 1.055) prevents the package from defaulting to STIX sans serif for math figures, leading instead to use of the figures in the Roman text font.